

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
## $n.burnin
## [1] 1000
##
## $n.draw
## [1] 1200
##
## $thin
## [1] 1
##
## $pp_cutoff
## [1] 0.5
##
## $prior_gamma
## [1] 0.1 0.1 0.1 0.1 0.1
##
## $K_values
## [1] "1 / Examined number of clusters: 2" "2 / Examined number of clusters: 3"
## [3] "3 / Examined number of clusters: 4" "4 / Examined number of clusters: 5"
## [5] "5 / Examined number of clusters: 6" "6 / Examined number of clusters: 7"
## [7] "7 / Examined number of clusters: 8" "8 / Examined number of clusters: 9"
## [9] "9 / Examined number of clusters: 10"
##
## $sdev
## [1] 0.01
##
## $beta.var.scale
## [1] 0.5
```

Summary Statistics

```
--- Summary for K = 1 ---
Z.ar (Location: fit[[1]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.6233  0.7067  0.7158  0.7151  0.7242  0.7550

beta.ar (Location: fit[[1]]$beta.ar):
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.09083 0.14083 0.13548 0.17417 0.95000
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Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[1]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09354 0.14500 0.13853 0.17833 0.75667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.100831 0.030833 1.000000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.076265 0.024167 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005833 0.015000 0.161686 0.119167 1.000000

--- Summary for K = 2 ---
Z.ar (Location: fit[[2]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.5042  0.5850  0.6033  0.5996  0.6175  0.6600

beta.ar (Location: fit[[2]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.04167 0.09167 0.09798 0.13667 0.87000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar (Location: fit[[2]]$gamma.ar):
  Modality 1 :

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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.04333	0.09500	0.10027	0.14167	0.68167

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0008333	0.0833333	0.3733881	0.8741667	1.0000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9158	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.003333	0.223283	0.262083	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3258	1.0000	0.6994	1.0000	1.0000

--- Summary for K = 3 ---

Z.ar (Location: fit[[3]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4808	0.5308	0.5433	0.5418	0.5542	0.5825

beta.ar (Location: fit[[3]]\$beta.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02333	0.05917	0.07241	0.10250	0.88667

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar (Location: fit[[3]]\$gamma.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02500	0.06167	0.07455	0.10667	0.57917

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.09583	0.57500	0.54213	1.00000	1.00000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1977	0.1183	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5275	1.0000	0.7379	1.0000	1.0000

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--- Summary for K = 4 ---
Z.ar (Location: fit[[4]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.4733  0.5350  0.5500  0.5509  0.5683  0.6242

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beta.ar (Location: fit[[4]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01417 0.04167 0.05810 0.08146 0.85083
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

```

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gamma.ar (Location: fit[[4]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01500 0.04500 0.05959 0.08500 0.61250
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.2141  0.4342  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  1.0000  1.0000  0.9997  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000  0.000  0.000  0.125  0.000  1.000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0125  0.4775  0.4605  0.7967  1.0000

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--- Summary for K = 5 ---
Z.ar (Location: fit[[5]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.4200  0.4683  0.4858  0.4838  0.4992  0.5550

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beta.ar (Location: fit[[5]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.01250 0.04375 0.05669 0.08167 0.82500
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

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```

Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1

gamma.ar (Location: fit[[5]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01333 0.04667 0.05912 0.08667 0.58500
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.1621  0.2825  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  1.0000  1.0000  0.9971  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.00000 0.00000 0.08257 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.08667 0.37417 0.37125 0.59167 1.00000

--- Summary for K = 6 ---
Z.ar (Location: fit[[6]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.8683  0.9808  0.9858  0.9843  0.9892  0.9975

beta.ar (Location: fit[[6]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01083 0.02667 0.04727 0.05833 0.86333
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1

gamma.ar (Location: fit[[6]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01167 0.02833 0.04778 0.06167 0.54917
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0001775 0.0000000 1.0000000
  Modality 3 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0001244	0.0000000	0.1950000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0002603	0.0000000	1.0000000

--- Summary for K = 7 ---

Z.ar (Location: fit[[7]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7558	0.9217	0.9425	0.9362	0.9583	0.9808

beta.ar (Location: fit[[7]]\$beta.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.008333	0.021667	0.043490	0.049167	0.868333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar (Location: fit[[7]]\$gamma.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.008333	0.023333	0.042701	0.051667	0.490000

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.005696	0.0000000	1.0000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.000124	0.0000000	1.0000000

--- Summary for K = 8 ---

Z.ar (Location: fit[[8]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9442	0.9758	0.9817	0.9804	0.9867	0.9975

beta.ar (Location: fit[[8]]\$beta.ar):

Modality 1 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 0.000000 0.006667 0.016667 0.030273 0.036667 0.839167

Modality 2 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 1 1 1 1 1 1

Modality 3 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 1 1 1 1 1 1

Modality 4 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 1 1 1 1 1 1

Modality 5 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 1 1 1 1 1 1

gamma.ar (Location: fit[[8]]\$gamma.ar):

Modality 1 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 0.000000 0.006667 0.018333 0.031225 0.038333 0.501667

Modality 2 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 0 0 0 0 0 0

Modality 3 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 0 0 0 0 0 0

Modality 4 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 0 0 0 0 0 0

Modality 5 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 0.0000000 0.0000000 0.0000000 0.0005526 0.0000000 1.0000000

--- Summary for K = 9 ---

Z.ar (Location: fit[[9]]\$Z.ar):

Min. 1st Qu. Median Mean 3rd Qu. Max.
 0.8708 0.9425 0.9517 0.9497 0.9600 0.9808

beta.ar (Location: fit[[9]]\$beta.ar):

Modality 1 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 0.000000 0.005833 0.014167 0.028190 0.032500 0.835833

Modality 2 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 1 1 1 1 1 1

Modality 3 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 1 1 1 1 1 1

Modality 4 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 1 1 1 1 1 1

Modality 5 :
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 1 1 1 1 1 1

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gamma.ar (Location: fit[[9]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.006667 0.015000 0.028371 0.035000 0.502500
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00e+00 0.00e+00 0.00e+00 4.01e-05 0.00e+00 1.00e+00
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.000000 0.000000 0.01388 0.000000 1.000000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
      0      0      0      0      0      0
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00e+00 0.00e+00 0.00e+00 8.28e-05 0.00e+00 1.00e+00

```